Biochemtools MCAT Formula Sheet

Free reference — every formula here has a full interactive calculator with worked steps at biochemtools.com

BIOCHEMISTRY

Peptide charge (Henderson–Hasselbalch, per group)

base: $+1/(1+10^{pH-pKa})$
acid: $-1/(1+10^{pKa-pH})$
pI = pH where net charge = 0

Michaelis–Menten

$v = V_{max}[S] / (K_m + [S])$
Lineweaver–Burk: $1/v = (K_m/V_{max})(1/[S]) + 1/V_{max}$
$K_m = [S]$ at $v = ½V_{max}$

Inhibition	K_m(app)	V_{max}(app)
Competitive	↑ (αK_m)	unchanged
Uncompetitive	↓ (K_m/α')	↓ (V_{max}/α')
Noncompetitive	unchanged	↓ (V_{max}/α')

MOLECULAR BIOLOGY

DNA melting temperature (T_m)

≤14 nt (Wallace): $T_m = 2(A+T) + 4(G+C)$
>14 nt: $T_m = 64.9 + 41(GC-16.4)/N$

Translation

Start: AUG (Met) Stop: UAA, UAG, UGA
DNA → mRNA: T → U

Glycolysis (per glucose)

Net: **+2 ATP, +2 NADH, 2 pyruvate**
Regulated steps: hexokinase, PFK-1 (committed), pyruvate kinase

GENETICS

Punnett ratios (complete dominance)

Monohybrid Aa×Aa: 3:1 phenotype
Dihybrid AaBb×AaBb: 9:3:3:1 phenotype
Test cross AaBb×aabb: 1:1:1:1

Hardy–Weinberg

$p + q = 1$ $p^2 + 2pq + q^2 = 1$
$p = (2AA+Aa)/2N$ $\chi^2 = \Sigma(O-E)^2/E$
χ^2 critical (df=1, α=0.05) = 3.84

CHEMISTRY

pH / pOH

Strong acid/base: [H+] or [OH–] = C
Weak acid: $K_a=x^2/(C-x)$; $x=(-K_a+\sqrt{(K_a{}^2+4K_aC)})/2$
pH + pOH = 14 (25°C)

Henderson–Hasselbalch (buffers)

pH = pKa + log([A–]/[HA])
ratio = 10^{pH-pKa}; buffers best at pKa ± 1

Gibbs free energy

$\Delta G = \Delta H - T\Delta S$ (spontaneous if $\Delta G<0$)
$\Delta G = \Delta G°' + RT \ln Q$ crossover $T = \Delta H/\Delta S$

Nernst equation

$E = E° - (RT/nF) \ln Q$
25°C shortcut: $E = E° - (0.0592/n) \log Q$
E=0 at equilibrium; F=96485 C/mol

Beer–Lambert

$A = \varepsilon bc$ $\%T = 100×10^{-A}$

Ideal gas law

PV = nRT R=0.08206 L·atm/mol·K
STP: 1 mol = 22.4 L (0°C, 1 atm)

SOLUTIONS & LAB PREP

Molarity & molar mass

moles = M × V(L) grams = moles × MW

Dilution

$C_1V_1 = C_2V_2$
Serial (n steps, fold F): conc = stock/F^n

PHYSIOLOGY

Osmotic pressure

Osmolarity = i × M $\pi = iMRT$
Blood ≈ 0.30 Osm (isotonic reference)

Cardiac output

CO = HR × SV
Fick: $CO = VO_2 / (CaO_2 - CvO_2)$
Normal resting CO ≈ 5 L/min

Renal clearance

$C_x = (U_x × V) / P_x$
FF = GFR / RPF
Inulin clearance = GFR (≈125 mL/min); PAH clearance ≈ RPF; normal FF ≈ 20%

biochemtools.com — 22 free interactive tools, no signup. Every formula above has a live calculator that shows the full worked steps.